

Networking Concepts for Cyber Security

IP Address

(Digital Address)

- A unique numerical label assigned to each device on a network.
- IPv4 → 192.168.1.1
- IPv6 → 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- Attackers trace IPs to launch attacks (e.g., DDoS, scanning).
- VPNs help hide IP addresses for privacy and protection.

DNS

(Domain Name System)

- The "phonebook of the internet" – translates human-friendly domain names into IP addresses.
- google.com → 142.250.190.14
- Vulnerable to DNS Spoofing/Poisoning (fake IP redirection).
- DNSSEC provides secure domain name resolution.

HTTP vs HTTPS

HTTP (HyperText Transfer Protocol):

- Transfers data without encryption.
- Vulnerable to sniffing and password leaks.

HTTPS (HTTP + SSL/TLS Encryption):

- Transfers data in encrypted form for secure communication.
- HTTPS is much harder for attackers to crack.
- Always use HTTPS websites for sensitive transactions.

VPN

(Virtual Private Network)

- Creates a secure encrypted tunnel for internet traffic while masking your IP address.
- Hides IP (ensures anonymity).
- Encrypts traffic (safe on public WiFi).
- Bypasses geo-restrictions.
- Prevents attackers from tracing your identity.
- Risk: Untrusted VPNs may log or track data.

Firewall Basics

- A security gatekeeper that monitors and controls network traffic based on allow/deny rules.
- Hardware Firewall
- Software Firewall
- Blocks unauthorized access.
- Prevents malware and hacking attempts.
- Works with IDS/IPS for advanced threat protection.